Experiment 3 – Cursor Based Row Processing in PL/pgSQL

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Experiment

Experiment 6: Implementing **implicit cursors, explicit cursors, and cursor attributes in PL/pgSQL** to process employee records row by row.

This experiment demonstrates how PostgreSQL handles query results internally and how programmers can manually control row-by-row processing using cursors

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Aim

To understand and implement **cursors in PL/pgSQL** for processing multiple rows from a database table one record at a time.

Objective

- To understand the concept of **cursors in PostgreSQL**.
- To implement **implicit cursors** for DML operations.
- To implement **explicit cursors** to fetch multiple records from a table.
- To use cursor attributes such as **FOUND, NOT FOUND, and row counters**.
- To process and display employee records using **RAISE NOTICE**.

Software Requirements

- Database: PostgreSQL
- Tool: iterm2

Practical / Experiment Steps

1. Create an **Employee table** and insert sample records.
2. Write a PL/pgSQL block to demonstrate **implicit cursor behavior** using an UPDATE statement.
3. Declare and use an **explicit cursor** to fetch employee records one by one.
4. Use a **loop structure** to process each row retrieved from the cursor.
5. Apply **cursor attributes** to control loop execution and count processed rows.
6. Display employee details using **RAISE NOTICE**.
7. Take screenshots of outputs for documentation.

Input / Output Details

Input

- ```
  emp_id INT
  name VARCHAR(50)
  salary INT
  ```
- Operations performed:
- Updating employee salary
- Fetching employee records using cursors
  ```

- Counting processed rows

Output

- Step 1: Display message indicating whether employee salary update was successful.
- Step 2: Display employee details fetched using an explicit cursor.
- Step 3: Display employee ID, name, and number of rows processed using cursor attributes.

SQL / PL/pgSQL Code

1.IMPLICIT CURSOR

```
DO $$
BEGIN
    UPDATE Employee
    SET salary = salary + 1000
    WHERE emp_id = 1;

    IF FOUND THEN
        RAISE NOTICE 'Employee salary updated successfully.';
    ELSE
        RAISE NOTICE 'Employee not found.';
    END IF;
END $$;
```

2.EXPLICIT CURSOR

```
DO $$
DECLARE
    emp_cursor CURSOR FOR
        SELECT emp_id, name, salary FROM Employee;

    emp_record RECORD;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO emp_record;
```

```
        EXIT WHEN NOT FOUND;

        RAISE NOTICE 'Employee ID: %, Name: %, Salary: %',
        emp_record.emp_id,
        emp_record.name,
        emp_record.salary;

    END LOOP;

    CLOSE emp_cursor;
END $$;
```

3.CURSOR ATTRIBUTE

```
DO $$
DECLARE
    emp_cursor CURSOR FOR
        SELECT emp_id, name FROM Employee;

    emp_record RECORD;
    row_counter INT := 0;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO emp_record;
        EXIT WHEN NOT FOUND;

        row_counter := row_counter + 1;

        RAISE NOTICE 'Employee ID: %, Name: %',
        emp_record.emp_id,
        emp_record.name;

        RAISE NOTICE 'Rows processed so far: %', row_counter;

    END LOOP;

    CLOSE emp_cursor;
END $$;
```

Screenshots

- **Screenshot 1:**Execution of implicit cursor program showing update result.

```
dinkygoyal=# DO $$
dinkygoyal$# BEGIN
dinkygoyal$#     UPDATE Employee
dinkygoyal$#     SET salary = salary + 1000
dinkygoyal$#     WHERE emp_id = 1;
dinkygoyal$#
dinkygoyal$#     IF FOUND THEN
dinkygoyal$#         RAISE NOTICE 'Employee salary updated successfully.';
dinkygoyal$#     ELSE
dinkygoyal$#         RAISE NOTICE 'Employee not found.';
dinkygoyal$#     END IF;
dinkygoyal$# END $$;
NOTICE:  Employee salary updated successfully.
DO
```

- **Screenshot 2:** Output displaying employee records fetched using explicit cursor.

```
dinkygoyal$# END $$;
NOTICE:  ID: 2, Name: Priya Verma, Salary: 45575
NOTICE:  ID: 3, Name: Rahul Mehta, Salary: 62000
NOTICE:  ID: 4, Name: Sneha Kapoor, Salary: 38745
NOTICE:  ID: 5, Name: Vikram Singh, Salary: 71000
NOTICE:  ID: 6, Name: Anjali Desai, Salary: 53211
NOTICE:  ID: 7, Name: Rohan Gupta, Salary: 48000
NOTICE:  ID: 8, Name: Neha Reddy, Salary: 39999
NOTICE:  ID: 9, Name: Karan Malhotra, Salary: 84500
NOTICE:  ID: 10, Name: Isha Nair, Salary: 36781
NOTICE:  ID: 11, Name: Arjun Patel, Salary: 90000
NOTICE:  ID: 12, Name: Meera Joshi, Salary: 42133
NOTICE:  ID: 13, Name: Siddharth Rao, Salary: 76000
NOTICE:  ID: 14, Name: Pooja Bansal, Salary: 58917
NOTICE:  ID: 15, Name: Manish Yadav, Salary: 64000
NOTICE:  ID: 1, Name: Amit Sharma, Salary: 51000
DO
```

- **Screenshot 3:** Output displaying employee details along with number of rows processed using cursor attributes.

```
dinkygoyal$# END $$;
NOTICE:  Employee ID: 2, Name: Priya Verma
NOTICE:  Rows processed so far: 1
NOTICE:  Employee ID: 3, Name: Rahul Mehta
NOTICE:  Rows processed so far: 2
NOTICE:  Employee ID: 4, Name: Sneha Kapoor
NOTICE:  Rows processed so far: 3
NOTICE:  Employee ID: 5, Name: Vikram Singh
NOTICE:  Rows processed so far: 4
NOTICE:  Employee ID: 6, Name: Anjali Desai
NOTICE:  Rows processed so far: 5
NOTICE:  Employee ID: 7, Name: Rohan Gupta
NOTICE:  Rows processed so far: 6
NOTICE:  Employee ID: 8, Name: Neha Reddy
NOTICE:  Rows processed so far: 7
NOTICE:  Employee ID: 9, Name: Karan Malhotra
NOTICE:  Rows processed so far: 8
NOTICE:  Employee ID: 10, Name: Isha Nair
NOTICE:  Rows processed so far: 9
NOTICE:  Employee ID: 11, Name: Arjun Patel
NOTICE:  Rows processed so far: 10
NOTICE:  Employee ID: 12, Name: Meera Joshi
NOTICE:  Rows processed so far: 11
NOTICE:  Employee ID: 13, Name: Siddharth Rao
NOTICE:  Rows processed so far: 12
NOTICE:  Employee ID: 14, Name: Pooja Bansal
NOTICE:  Rows processed so far: 13
NOTICE:  Employee ID: 15, Name: Manish Yadav
NOTICE:  Rows processed so far: 14
NOTICE:  Employee ID: 1, Name: Amit Sharma
NOTICE:  Rows processed so far: 15
DO
```

Learning Outcome

- Learned the concept and importance of **cursors in PL/pgSQL**.
- Understood the difference between **implicit and explicit cursors**.
- Learned how to **fetch records one by one from a table**.

- Learned how to use **cursor attributes such as FOUND and NOT FOUND**.
- Gained practical knowledge of **row-by-row data processing in PostgreSQL**.